Analysis of an algorithm usually deals with the running and execution time of various operations involved. The running time of an operation is calculated as how many programming instructions executed per operation.

Analysis of algorithms and performance evaluation can be divided into two different phases:

1. **A Priori estimates:** This is a theoretical performance analysis of an algorithm. Efficiency of an algorithm is measured by assuming the external factors.

2. **A Posteriori testing:** This is called performance measurement. In this analysis, actual statistics like running time and required for the algorithm executions are collected.

An estimation of the time and space complexities of an algorithm for varying input sizes is called algorithm analysis.

4.2 Complexity of an Algorithm

Suppose A is an algorithm and n is the size of input data, the time and space used by the algorithm A are the two main factors, which decide the efficiency of A.

Time Factor -Time is measured by counting the number of key operations like comparisons in the sorting algorithm.

Space Factor - Space is measured by the maximum memory space required by the algorithm.

The complexity of an algorithm f (n) gives the running time and/or the storage space required by the algorithm in terms of n as the size of input data.

4.2.1 Time Complexity

The Time complexity of an algorithm is given by the number of steps taken by the algorithm to complete the process.

4.2.2. Space Complexity

Space complexity of an algorithm is the amount of memory required to run to its completion. The space required by an algorithm is equal to the sum of the following two components:

A fixed part is defined as the total space required to store certain data and variables for an algorithm. For example, simple variables and constants used in an algorithm.

A variable part is defined as the total space required by variables, which sizes depends on the problem and its iteration. For example: recursion used to calculate factorial of a given value n.

4.3 Efficiency of an algorithm

Computer resources are limited that should be utilized efficiently. The efficiency of an algorithm is defined as the number of computational resources used by the algorithm. An algorithm must be analyzed to determine its resource usage. The efficiency of an algorithm can be measured based on the usage of different resources.

For maximum efficiency of algorithm we wish to minimize resource usage. The important resources such as time and space complexity cannot be compared directly,

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